

Chapter 11

Localized charges: the multipole expansion

11.1 The multipole expansion

For a localized charge distribution, we want to organize the potential far away as a systematic expansion. Start from the Taylor expansion of a function of $\vec{x}'$ about the origin

$$f(\vec{x}') = f(0) + \frac{\partial f}{\partial x'^i} \Big|_0 x'^i + \frac{1}{2!} \frac{\partial^2 f}{\partial x'^i \partial x'^j} \Big|_0 x'^i x'^j + \dots = \sum_{n=0}^{\infty} \frac{(\vec{x}' \cdot \vec{\nabla}')^n}{n!} f(\vec{x}') \Big|_{\vec{x}'=0} \quad (11.1)$$

Take

$$f(\vec{x}') = \frac{1}{|\vec{x} - \vec{x}'|} \quad (11.2)$$

which gives, writing $x \equiv |\vec{x}|$,

$$f(0) = \frac{1}{x} \quad , \quad \frac{\partial f}{\partial x'^i} \Big|_0 = \frac{x^i}{x^3} \quad , \quad \frac{\partial^2 f}{\partial x'^i \partial x'^j} \Big|_0 = \frac{3 x^i x^j - x^2 \delta^{ij}}{x^5} \quad (11.3)$$

With the boundary condition $\phi \rightarrow 0$ at infinity, the potential is then

$$\begin{aligned} \phi(\vec{x}) &= \int \rho(\vec{x}') \frac{1}{|\vec{x} - \vec{x}'|} d^3V' \\ &= \frac{Q}{x} + \frac{\vec{p} \cdot \vec{x}}{x^3} + \frac{1}{2} \frac{3 x^i x^j - x^2 \delta^{ij}}{x^5} \int_{V'} \rho(\vec{x}') x'^i x'^j dV' + \dots \end{aligned} \quad (11.4)$$

The last term can be recast as

$$\frac{1}{6} \frac{3 x^i x^j - x^2 \delta^{ij}}{x^5} Q_{ij} \quad (11.5)$$

(the δ^{ij} piece added to the integrand drops against the traceless prefactor), where

$$p^i = \int \rho(\vec{x}') x'^i dV' \quad (11.6)$$

is the *dipole moment* and

$$Q_{ij} = \int \rho(\vec{x}') (3 x'^i x'^j - (x')^2 \delta^{ij}) dV' \quad (11.7)$$

is the *quadrupole moment*. Note that

$$Q_{11} + Q_{22} + Q_{33} = 0 \quad , \quad Q_{ij} = Q_{ji} \Rightarrow 5 \text{ independent components} \quad (11.8)$$

11.2 Expansion in spherical coordinates

Alternatively, we can expand in spherical coordinates. Using the spherical harmonic expansion of $1/|\vec{x} - \vec{x}'|$,

$$\phi(\vec{x}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{4\pi}{2\ell+1} \frac{1}{r^{\ell+1}} Y_{\ell m}(\theta, \varphi) \int \rho(\vec{x}') r'^{\ell} Y_{\ell m}^*(\theta', \varphi') dV' \quad (11.9)$$

which leads us to define the *multipole moments*

$$q_{\ell m} \equiv \int Y_{\ell m}(\theta', \varphi') r'^{\ell} \rho(\vec{x}') d^3V' \quad (11.10)$$

Counting components at each order: $\ell = 0$ is the monopole (1 component); $\ell = 1$ is the dipole (3 components, $m = -1, 0, 1$); $\ell = 2$ is the quadrupole (5 components, $m = -2, \dots, 2$) – matching the Cartesian counting. The first few moments are related to the Cartesian ones by

$$q_{00}^* = \frac{Q}{\sqrt{4\pi}} \quad (11.11)$$

$$q_{11}^* = -\sqrt{\frac{3}{8\pi}} (p^x - i p^y) = -q_{1,-1} \quad , \quad q_{10}^* = \sqrt{\frac{3}{4\pi}} p^z \quad [?] \quad (11.12)$$

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$$q_{22}^* = \frac{1}{12} \sqrt{\frac{15}{2\pi}} (Q_{11} - 2i Q_{12} - Q_{22}) = q_{2,-2} \quad (11.13)$$

$$q_{21}^* = -\frac{1}{6} \sqrt{\frac{15}{2\pi}} (Q_{13} - i Q_{23}) = -q_{2,-1} \quad , \quad q_{20}^* = \frac{1}{4} \sqrt{\frac{5}{\pi}} Q_{33} \quad (11.14)$$

11.3 Interaction of localized charges with external fields

Consider next a localized charge distribution sitting in an external potential ϕ_{ext} . Expanding about the origin,

$$\phi_{\text{ext}}(\vec{x}) = \phi_{\text{ext}}(0) + \vec{x} \cdot \vec{\nabla} \phi_{\text{ext}} \Big|_0 + \frac{1}{2} x^i x^j \partial_i \partial_j \phi_{\text{ext}} \Big|_0 + \cdots \quad (11.15)$$

with

$$\vec{\nabla} \phi_{\text{ext}} \Big|_0 = -\vec{E}_{\text{ext}}(0) \quad , \quad \partial_i \partial_j \phi_{\text{ext}} \Big|_0 = -\partial_i E_{\text{ext}}^j \Big|_0 \quad (11.16)$$

This gives

$$\phi_{\text{ext}}(\vec{x}) = \phi_{\text{ext}}(0) - \vec{x} \cdot \vec{E}_{\text{ext}}(0) - \frac{1}{6} (3 x^i x^j - x^2 \delta^{ij}) \partial_i E_{\text{ext}}^j \Big|_0 + \cdots \quad (11.17)$$

where we used $\vec{\nabla} \cdot \vec{E}_{\text{ext}} = 0$ in the region of interest to add the $x^2 \delta^{ij}$ piece for free. The interaction energy is then

$$W = \int \rho(\vec{x}) \phi_{\text{ext}}(\vec{x}) dV = q \phi_{\text{ext}}(0) - \vec{p} \cdot \vec{E}_{\text{ext}}(0) - \frac{1}{6} Q_{ij} \partial_i E_{\text{ext}}^j \Big|_0 \quad (11.18)$$

11.4 Dipole–dipole energy

A dipole $\vec{p}$ at the origin has potential

$$\phi(\vec{x}) = \frac{\vec{p} \cdot \vec{x}}{x^3} \quad (11.19)$$

so its electric field is

$$\begin{aligned} \vec{E} = -\vec{\nabla} \phi &= -\frac{\vec{\nabla} (\vec{p} \cdot \vec{x})}{x^3} - (\vec{p} \cdot \vec{x}) \vec{\nabla} \left(\frac{1}{x^3} \right) \\ &= -\frac{\vec{p} - 3 \frac{\vec{x}}{x} (\frac{\vec{x}}{x} \cdot \vec{p})}{x^3} \end{aligned} \quad (11.20)$$

Using (11.18), the energy of a second dipole $\vec{p}_2$ in the field of $\vec{p}_1$, with $\vec{x}$ the separation vector as depicted in the Figure, is

$$W_{12} = -\vec{p}_2 \cdot \vec{E}_1 = \frac{\vec{p}_1 \cdot \vec{p}_2 - 3 \left(\vec{p}_1 \cdot \frac{\vec{x}}{x} \right) \left(\vec{p}_2 \cdot \frac{\vec{x}}{x} \right)}{x^3} \quad (11.21)$$

Defining $W \equiv p_1 p_2 / x^3$, the characteristic configurations give

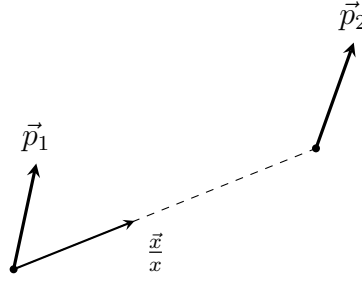


Figure 11.1: Two dipoles $\vec{p}_1$ and $\vec{p}_2$ separated by $\vec{x}$, with unit vector $\vec{x}/x$ along the separation.

$$W_{12} = -2W \quad (\rightarrow \rightarrow) \quad , \quad -W \quad (\uparrow \downarrow) \quad , \quad +W \quad (\uparrow \uparrow) \quad , \quad +2W \quad (\rightarrow \leftarrow) \quad (11.22)$$

where the arrows denote the dipole orientations relative to the separation axis, taken horizontal. (Including the location of the dipole itself, the exact dipole field is

$$\vec{E}(\vec{r}) = \frac{3 \frac{\vec{x}}{x} (\vec{p} \cdot \frac{\vec{x}}{x}) - \vec{p}}{x^3} - \frac{4\pi}{3} \vec{p} \delta^3(\vec{x} - \vec{x}_0) \quad (11.23)$$

for a dipole located at $\vec{x}_0$.)

11.5 Examples

As a first example, take the linear quadrupole

$$\rho(\vec{x}) = q \delta(x) \delta(y) (\delta(z - a) - 2 \delta(z) + \delta(z + a)) \quad (11.24)$$

depicted in the Figure Using (11.7), this gives

$$Q_{11} = Q_{22} = -2 a^2 q \quad , \quad Q_{33} = 4 a^2 q \quad (11.25)$$

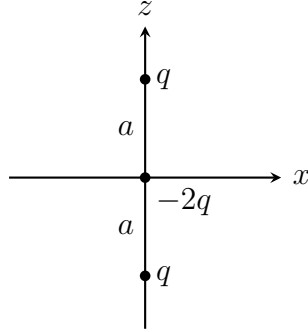


Figure 11.2: Linear quadrupole: charges q at $z = \pm a$ and $-2q$ at the origin.

Next, take the planar quadrupole

$$\rho(\vec{x}) = q \delta(z) [\delta(x) (\delta(y - a) + \delta(y + a)) - \delta(y) (\delta(x - a) + \delta(x + a))] \quad (11.26)$$

depicted in the Figure which gives

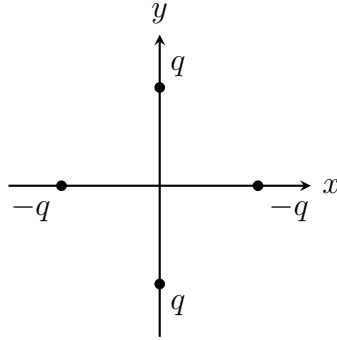


Figure 11.3: Planar quadrupole: charges q at $y = \pm a$ and $-q$ at $x = \pm a$.

$$Q_{11} = 6 a^2 q \text{ [?]} \quad , \quad Q_{22} = -6 a^2 q \quad (11.27)$$

and the spherical moments

$$q_{22} = q_{2,-2} = \sqrt{\frac{15}{2\pi}} q a^2 \quad (11.28)$$

Finally, let us find *all* the multipoles of the planar quadrupole. In spherical coordinates,

$$\rho(\vec{x}) = q \frac{\delta(r-a)}{r^2} \frac{\delta(\theta - \frac{\pi}{2})}{\sin \theta} \left[\delta(\varphi) - \delta\left(\varphi - \frac{\pi}{2}\right) + \delta(\varphi - \pi) - \delta\left(\varphi - \frac{3\pi}{2}\right) \right] \quad (11.29)$$

so that (11.10) gives

$$q_{\ell m} = q a^\ell \sum_{n=0}^3 (-1)^n Y_{\ell m}\left(\frac{\pi}{2}, \frac{n\pi}{2}\right) \quad (11.30)$$

Using

$$Y_{\ell m}\left(\frac{\pi}{2}, \frac{n\pi}{2}\right) = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}} P_\ell^m(0) e^{i m n \frac{\pi}{2}} \quad (11.31)$$

we get

$$q_{\ell m} = q a^\ell \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}} P_\ell^m(0) \sum_{n=0}^3 (-1)^n e^{i m n \frac{\pi}{2}} \quad (11.32)$$

The sum over n collapses onto

$$q_{\ell, -2} = 4 q a^\ell \sqrt{\frac{(2\ell+1)(\ell+2)!}{4\pi(\ell-2)!}} P_\ell^{-2}(0) \quad , \quad \ell = 2, 4, \dots \quad (11.33)$$

where

$$P_\ell^m(0) = \frac{2^m}{\sqrt{\pi}} \cos\left(\frac{\pi}{2}(\ell+m)\right) \frac{\Gamma\left(\frac{\ell}{2} + \frac{m}{2} + \frac{1}{2}\right)}{\Gamma\left(\frac{\ell}{2} - \frac{m}{2} + 1\right)} \quad [?] \quad (11.34)$$